

Notes from Analysis in the Marginalia

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The following is an enumeration of unrelated results from over the year, most of which pertaining to pieces of analysis with a focus on closed forms for various limits, problems, sums, integrals, and processes. Nearly all are curiosities, but may hopefully inspire more productive findings.

Taking a page out of Raabe's Formula;

$$\ln \left| \frac{\zeta(s)}{\zeta(1-s)} \right| = \ln \left| 2^s \pi^{(s-1)} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \right| \rightarrow \int_a^{(a+1)} \ln \left| \frac{\zeta(s)}{\zeta(1-s)} \right| ds =$$

$$\int_a^{a+1} \ln \left(\left| 2^s \pi^{(s-1)} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \right| \right) ds = \ln \left(\frac{2\pi e}{a} \right) a + \frac{2}{\pi} \sum_{n=0}^{\infty} \frac{\sin((2n+1)\pi a)}{(2n+1)^2}$$

In the case where $a = \frac{2k+1}{2}$ the above sum produces Catalan's constant G ;

$$= \ln \left(\frac{4\pi e}{2k+1} \right) \frac{2k+1}{2} + \frac{(-1)^k 2}{\pi} G$$

What is the average digit of a variable? The extraction of the n th digit is given by;

$$\text{floor}(X \cdot 10^n) - 10 \text{floor} \left(\frac{X}{10} \cdot 10^n \right)$$

$$\int_0^1 \left(\text{floor}(x \cdot 10^n) - 10 \text{floor} \left(\frac{x}{10} \cdot 10^n \right) \right) dx = \frac{1}{10^n} \left(\frac{10^n(10^n-1)}{2} - 100 \frac{\left(\frac{10^n}{10}\right) \left(\frac{10^n}{10}-1\right)}{2} \right) = 4.5 \forall n \geq 1$$

$$\int_0^{\frac{\pi}{2}} \sin(x) dx = 1$$

$$\int_0^{\frac{\pi}{2}} \sin(\sin(x)) dx = \Im \int_0^{\frac{\pi}{2}} e^{i \sin(x)} dx = \frac{\sqrt{\pi}}{2} \sum_{n=0}^{\infty} \frac{\sin\left(\frac{\pi}{2}n\right)}{n!} \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}+1\right)} = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot n!^2 \cdot 4^n}{(2n+1)!^2} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!!^2}$$

The Normal approximation to the Binomial distribution is intriguing from the formula perspective;

$$e^{-\frac{1}{2}x^2} \sim p^{\left(\sqrt{Np(1-p)}x + Np\right)} \cdot (1-p)^{N-\left(\sqrt{Np(1-p)}x + Np\right)} \cdot \text{nCr}\left(N, \sqrt{Np(1-p)}x + Np\right) \sqrt{2\pi Np(1-p)}$$

Inserting $p = 1/2$ for simplification makes for an intriguing family of limits;

$$\lim_{N \rightarrow \infty} N \cdot 16^{-N^2} \binom{4N^2}{Nx + 2N^2} = \frac{e^{-\frac{1}{2}x^2}}{\sqrt{2\pi}}$$

Stirling's approximation can aid in proving such results as;

$$\lim_{N \rightarrow \infty} \frac{1}{2^N} \text{nCr} \left(N, \frac{\sqrt{N}}{2}x + \frac{N}{2} \right) \frac{\sqrt{N}}{2} \Rightarrow N \rightarrow 4N^2 \Rightarrow \text{Stirling} \Rightarrow 16^{-N^2} \cdot \frac{2N(4N)^{\binom{4N^2}{2N^2-Nx}} \left(\frac{2N^2-Nx}{2N^2+Nx} \right)^{\binom{Nx}{2N^2}}}{\sqrt{4N^2-x^2} (4N^2-x^2)^{\binom{2N^2}{2N^2}}}$$

$$= \lim_{N \rightarrow \infty} \frac{2^{4N^2}}{4^{2N^2}} \frac{1}{\left(1 - \frac{x^2}{4N^2}\right)^{N^2}} \cdot \left(\left(1 - \frac{2x}{2N+x}\right)^N \right)^x \cdot \frac{2N}{\sqrt{4N^2 - x^2}} \cdot \left(16^{-N^2} \cdot 2^{4N^2}\right) = 1 \cdot \frac{1}{\left(e^{-\frac{x^2}{4}}\right)^2} \cdot (e^{-x})^x \cdot 1 \cdot 1$$

If $A = uv^T$, then

$$A^2 = uv^T uv^T = (u \cdot v) (uv^T) = (u \cdot v) (A)$$

Giving rise to an infinite family of interesting duplication matrices;

$$\begin{pmatrix} 3 & 6 \\ 4 & 8 \end{pmatrix}^2 = \begin{pmatrix} 33 & 66 \\ 44 & 88 \end{pmatrix}$$

$$\begin{pmatrix} 45 & 40 \\ 63 & 56 \end{pmatrix}^2 = \begin{pmatrix} 4545 & 4040 \\ 6363 & 5656 \end{pmatrix}$$

$$\begin{pmatrix} 14 & 12 & 18 \\ 70 & 60 & 90 \\ 21 & 18 & 27 \end{pmatrix}^2 = \begin{pmatrix} 1414 & 1212 & 1818 \\ 7070 & 6060 & 9090 \\ 2121 & 1818 & 2727 \end{pmatrix}$$

Define the ‘Rubiks Cube Average’, used in speedcubing competitions, as the arithmetic average of the given entries after excluding the maximum and minimum. This quantity is explicitly given by;

$$\text{Rubiks}(\vec{v}) = \lim_{N \rightarrow \infty} \frac{1}{\text{len}(\vec{v}) - 2} \left(\left(\sum_i v_i \right) - \left(\sum_i v_i^{-N} \right)^{\frac{1}{-N}} - \left(\sum_i v_i^N \right)^{\frac{1}{N}} \right)$$

Pleasant patterns in the generating functions for Dirichlet Series;

$$\frac{\pi 0 \cot(\pi 0) - \pi x \cot(\pi x)}{2} = \sum_{n=1}^{\infty} x^{2n} \zeta(2n)$$

$$\frac{\sec\left(\frac{\pi}{2} 0\right) \frac{\pi}{2} 0 - \sec\left(\frac{\pi x}{2}\right) \frac{\pi x}{2}}{2} = - \sum_{n=0}^{\infty} x^{2n+1} \beta(2n+1)$$

$$\frac{\pi 0 \csc(\pi 0) - \pi x \csc(\pi x)}{2} = - \sum_{n=1}^{\infty} x^{2n} \eta(2n)$$

$$\frac{\pi x}{6} \left(\frac{\sin(2\pi x)}{\cos(2\pi x) - 1} + \frac{\sin(\pi x) + \sqrt{3} \sinh(\sqrt{3}\pi x)}{\cos(\pi x) - \cosh(\sqrt{3}\pi x)} \right) + \frac{1}{2} = \sum_{n=1}^{\infty} \zeta(6n) x^{6n}$$

Approximations for $\pi/12$ from rational functions of sine and cosine;

$$\frac{\frac{1}{20} - \frac{1}{240\sqrt{2}} + \frac{25}{32} \frac{\sqrt{3}-1}{\sqrt{2}}}{1 + \frac{3}{8} \frac{1+\sqrt{3}}{\sqrt{2}}} - \frac{\pi}{12} \approx 10^{-10}$$

$$\frac{1}{240} \left(-338 - 172\sqrt{2} - 7\sqrt{3} + 244\sqrt{6} \right) - \frac{\pi^3}{12^3} \approx 10^{-8}$$

And other aesthetic approximations;

$$\frac{1000001^6}{10^{30} \cdot 1000007} \approx 1$$

$$10e^{\frac{1}{5}} - \frac{5}{3}e^{\frac{2}{5}} - \frac{10}{3}e^{-\frac{1}{5}} = 6.9986$$

$$\frac{20}{9} \left(\sqrt{3} - 1 \right) - \frac{5\pi}{36} \left(\sqrt{3} + 1 \right) = \left(\sinh\left(\frac{\pi}{12}\right) - \frac{1}{9} \right) 2\sqrt{2} + \sim 10^{-6}$$

$$\frac{11 \left(\frac{\sqrt{3}-1}{2\sqrt{2}} \right) + 1}{\pi} - \left(\frac{\sqrt{3}+1}{2\sqrt{2}} \right) - \frac{\cosh\left(\frac{\pi}{12}\right)}{4} \sim -0.00000108$$

$$\left(23 + \frac{5}{9}\right) \cosh\left(\frac{\pi}{12}\right) - \left(18 + \frac{7}{9}\right) \cosh\left(\frac{\pi}{6}\right) + 6 \cosh\left(\frac{\pi}{4}\right) - \left(\frac{7}{9}\right) \cosh\left(\frac{\pi}{3}\right) = 5 \frac{\sqrt{3} + 1}{\sqrt{2}} + \sim 2 \cdot 10^{-7}$$

$$\frac{-\ln\left(\frac{38086}{9999}\right) + 18 \ln(10)}{\sqrt{163}} - \pi \sim 8 \cdot 10^{-10}$$

A nice coupling of Lambert Series & Laurent Series;

$$\sum_{n=1}^{\infty} z^n \sigma(n) = \sum_{n=1}^{\infty} \frac{z^n}{1-z^n} \rightarrow \sigma(n) = \frac{1}{2\pi i} \sum_{k=0}^{n-1} \oint_{|z|=r<1} \frac{1}{1-z^{n-k}} \frac{1}{z^{(k+1)}} dz$$

As is to be expected, but is nonetheless interesting, this contour integral (about the origin) on its own, below, represents $a \mid b-1$

$$\frac{1}{2\pi i} \oint \frac{1}{1-z^a} \cdot \frac{1}{z^b} dz = \sum_{k=0}^{a-1} \frac{1}{\left(a \left(e^{\frac{2\pi i k}{a}}\right)^{a-1}\right) \left(\left(e^{\frac{2\pi i k}{a}}\right)^b\right)} = \sum_{k=0}^{a-1} \frac{1}{a} \cdot e^{\frac{2\pi i k(1-b)}{a}} = a \mid b-1$$

$$\sigma(n) = \sum_{k=0}^{n-1} \frac{1}{2\pi} \int_0^{2\pi} \frac{r^k \cos(k\theta) - r^n \cos(n\theta)}{r^{2k} + r^{2n} - 2r^{(n+k)} (\cos(n\theta - k\theta))} d\theta$$

And hence we have an explicit form for the count-of-divisors function!

Dirichlet Series have an interesting interplay with floor-function based recursion;

$$a_n = a_{\lceil \frac{n}{k_1} \rceil} + a_{\lceil \frac{n}{k_2} \rceil} + a_{\lceil \frac{n}{k_3} \rceil} \dots$$

We define b_n and note that the floor functions act to produce;

$$b_n = a_{n+1} - a_n = 1_{\{k_1 \setminus n+1\}} b_{\frac{n}{k_1}} + 1_{\{k_2 \setminus n+1\}} b_{\frac{n}{k_2}} + 1_{\{k_3 \setminus n+1\}} b_{\frac{n}{k_3}} \dots$$

And hence their Dirichlet generating series;

$$B(s) = \sum_{n=1}^{\infty} \frac{b_n}{n^s} = b_1 + \sum_{n=2}^{\infty} \frac{b_{\frac{n}{k_1}} \{k_1 \setminus n\}}{n^s} + \sum_{n=2}^{\infty} \frac{b_{\frac{n}{k_2}} \{k_2 \setminus n\}}{n^s} + \sum_{n=2}^{\infty} \frac{b_{\frac{n}{k_3}} \{k_3 \setminus n\}}{n^s} \dots$$

$$b_1 + \sum_{j=1}^{\infty} \frac{b_j}{(k_1 j)^s} + \sum_{j=1}^{\infty} \frac{b_j}{(k_2 j)^s} + \sum_{j=1}^{\infty} \frac{b_j}{(k_3 j)^s} \Rightarrow B(s) = \frac{c}{1 - k_1^{-s} - k_2^{-s} - k_3^{-s} \dots}$$

And with Perron we can now recover the highly non-elementary series in explicit form;

$$\frac{b_N}{2} + \sum_{n=1}^{N-1} b_n = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} B(s) \frac{N^s}{s} ds = \frac{b_N}{2} + a_N - a_1$$

Such investigations are motivated by the Erdos Problem with 2,3, and 6 for k_1, k_2, k_3 .

It is well known that the Laplacian is invariant under Orthogonal transformations;

$$\hat{x} = Qx \rightarrow \frac{\partial}{\partial \hat{x}_i} = \sum_j Q_{ij} \frac{\partial}{\partial x_j}$$

$$\Delta_{\hat{x}} = \sum_k \left(\frac{\partial}{\partial \hat{x}_k} \right) \left(\frac{\partial}{\partial \hat{x}_k} \right) = \sum_k \sum_a \sum_b Q_{ak} Q_{bk} \frac{\partial}{\partial x_k} \frac{\partial}{\partial x_k} = \sum_k \frac{\partial^2}{\partial x_k^2} (Q^T Q) = \Delta_x$$

Interestingly however the divergence is invariant under Stochastic matrices;

$$\hat{x} = Sx \rightarrow \frac{\partial}{\partial \hat{x}_i} = \sum_j S_{ij} \frac{\partial}{\partial x_j} \Rightarrow \Delta \cdot = \sum_k \left(\frac{\partial}{\partial \hat{x}_k} \right) = \sum_k \sum_a S_{ak} \frac{\partial}{\partial x_k} = \sum_k \frac{\partial}{\partial x_k} (1) = \Delta \cdot$$

The function $f(x) = \sum_{n=0}^{\infty} x^{2^n}$ satisfies the expansion;

$$\exp\left(\sum_{k=1}^{\infty} \frac{(-1)^k}{k} f(x^k)\right) = 1 - x$$

Neglecting the boundedness requirements of the Mittag Leffler expansion produces approximations quite conveniently. Γ is of course unbounded outside of the meromorphic region, and yet its values on the negative axis are nonetheless *estimated* by;

$$1 + \sum_{n=1}^{\infty} \frac{(-1)^{(n+1)}}{(n-1)!} \left(\frac{1}{x-1+n} - \frac{1}{n} \right)$$

Similarly, taking $1/\zeta$ and using standard results concerning the trivial zeros we have, for negative x , the approximation;

$$-2 + \sum_{n=1}^{\infty} \frac{1}{\frac{(-1)^n (2n)! \zeta(2n+1)}{2 \cdot (2\pi)^{2n}}} \left(\frac{1}{z - (-2n)} + \frac{1}{(-2n)} \right)$$

By inspection we have;

$$\sum_{n=1}^{\infty} \mathcal{L}_{f(t)}(\ln(n)) x^n = \int_0^{\infty} \left(\sum_{n=1}^{\infty} \frac{x^n}{n^t} \right) f(t) dt = \int_0^{\infty} Li_t(x) f(t) dt$$

Which is interesting. For $x = 1, -1$, this collapses to the zeta or eta function. Trivially we also have;

$$\sum_{n=1}^{\infty} \mathcal{L} \left\{ \frac{\alpha(t)}{\zeta(t)} \right\} (\ln(n)) = \sum_{n=1}^{\infty} \int_0^{\infty} e^{-\ln(n)t} \frac{\alpha(t)}{\zeta(t)} dt = \int_0^{\infty} \alpha(t) dt$$

Setting;

$$f(s) = \int_0^{\infty} e^{-st} Li_t(q) dt$$

We have the convenient;

$$\sum_{n=1}^{\infty} f(\ln(n)) q^n \Lambda(n) = \frac{q}{1-q}$$

For $\Lambda(0) = 0$, the Von-Mandgoldt function. This identity is peculiar as it insinuates that $f(\ln(n))\Lambda(n) = 1$, giving the similar series of $\sum_{n=1}^{\infty} q^n = \frac{q}{1-q}$.

A mildly interesting formula for $\ln(x)$;

$$\sum_{m=1}^{\infty} \left(\left(\sum_{k=0}^{x-1} \frac{1}{xm + xX + k - x} \right) - \frac{1}{m} \right) - \sum_{m=1}^{\infty} \left(\frac{1}{m + xX - 1} - \frac{1}{m} \right)$$

Or simpler;

$$\ln(x) = \sum_{m=1}^{\infty} \left(\left(\sum_{k=0}^{x-1} \frac{1}{xm + k} \right) - \frac{1}{m} \right) + \sum_{k=1}^{x-1} \frac{1}{k}$$

Giving Euler–Mascheroni as;

$$\lim_{N \rightarrow \infty} \sum_{m=1}^N \frac{1}{m} - \sum_{k=0}^{N-1} \sum_{m=1}^N \left(\frac{1}{Nm + k} \right) = \gamma$$

The function;

$$K(x) = -\ln\left(1 + \sqrt{1 - \frac{(c+2x)^2}{4}}\right) - \ln\left(\sqrt{\frac{4}{(c+2x)^2}}\right) + \ln\left(\frac{2}{c} + \frac{2}{c}\sqrt{1 - \frac{c^2}{4}}\right) + \sqrt{1 - \frac{(c+2x)^2}{4}} - \sqrt{1 - \frac{c^2}{4}} + c_2$$

Is such that the classical rate of curvature, or derivative rather, is the same as the radius of curvature (up to sign). Satisfying therefore, the elementary ODE;

$$|y'| = \left| \frac{\left(1 + y'(x)^2\right)^{\frac{3}{2}}}{y''(x)} \right|$$

An elegant association between Ramanujan's master theorem and Lindelöf summation;

$$\begin{aligned} \mathcal{M}\{f(x)\} &= \Gamma(s) \phi(-s) \rightarrow \sum_{k=0}^{\infty} \frac{\phi(k)}{k!} (-x)^k = \mathcal{M}^{-1}\{\Gamma(s) \phi(-s)\} \\ \sum_{k=0}^{\infty} \phi(k) (-1)^k &= \mathcal{M}^{-1}\{\Gamma(s) \phi(-s) \cdot (-s)!\} [1] \\ \sum_{k=0}^{\infty} \phi(k) (-1)^k &= \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \phi(-s) \frac{\pi}{\sin(\pi s)} ds = \Re \frac{1}{2} \int_{-\infty}^{\infty} \phi\left(-\left(\frac{1}{2} + it\right)\right) \frac{1}{\sin\left(\pi\left(\frac{1}{2} + it\right)\right)} dt \\ &= \frac{1}{2} \int_{-\infty}^{\infty} \phi\left(-\frac{1}{2} + it\right) \operatorname{sech}(\pi t) dt \end{aligned}$$

Boundary of a boundary is Null;

$$\oint_{\partial\partial S} u dr = \iint_{\partial S} (\nabla \times u) \cdot n dS = \iiint_S \nabla \cdot (\nabla \times u) dV = 0$$

The convergence of the expansion of the Riemann-Siegel theta function directly implies the quite accurate;

$$\begin{aligned} \sum_{n=1}^x \left(\arctan\left(\frac{2X}{4n+1}\right) \right) &\sim \frac{X}{2} (\ln(x) + g) - \left(\frac{X}{2} \ln\left(\frac{X}{2\pi}\right) - \frac{X}{2} - \frac{\pi}{8} + \frac{g + \ln(\pi)}{2} X + \arctan(2X) \right) \\ \sum_{n=1}^x C\left(\frac{\pi(4n+1)}{2}\right) &\sim \frac{\ln(x) + g}{2\pi} \end{aligned}$$

What about $\exp(1/n)$ or $\arctan(1/n)$ etc. Each of these have interesting asymptotics.

Differential equations of the form below have a trivial reduction of order scheme;

$$F_x + F_y y' + F_{y'} y'' + F_{y''} y''' \dots = 0 = \frac{d}{dx} F(x, y, y' \dots) \rightarrow F = c$$

In much the same way as exact equations, ODEs with the correct structure can be swiftly dealt with.

The following product expansion is a generalization of a problem taken from Mu Prime Math;

$$\ln(a) = (a-1) \prod_{k=1}^{\infty} \left(\frac{2}{a^{\frac{1}{2^k}} + 1} \right) \rightarrow \ln(\ln(x)) = \ln(x-1) - \sum_{k=1}^{\infty} \ln\left(\frac{x^{\frac{1}{2^k}} + 1}{2}\right)$$

A relaxing Central Binomial identity;

$$\sum_{n=1}^{\infty} \frac{n}{2n-1} \frac{1}{\operatorname{nCr}(4n, 2n)} = \frac{180 + 25\sqrt{3}\pi + 297\sqrt{5} \ln\left(\frac{3+\sqrt{5}}{2}\right)}{5400}$$

This is special case of a more general family of central binomial identities;

$$\sum_{k=1}^{\infty} \frac{1}{(2N-3+4k)} \frac{k}{\operatorname{nCr}(2N-4+4k, N-2+2k)} = \int_0^1 \frac{x^N (1-x)^N}{\left(1-x^2(1-x)^2\right)^2} dx$$

Slicing an Apple: Which linear slice, $ax+b$, removes a predetermined volume from the upper hemisphere of a unit-circular apple at the origin while leaving the minimal exposed area? The intersections at the circle occur at;

$$x_0 = \frac{-ab - \sqrt{a^2 - b^2 + 1}}{a^2 + 1} ; x_1 = \frac{-ab + \sqrt{a^2 - b^2 + 1}}{a^2 + 1}$$

With arclength of;

$$\int_{x_0}^{x_1} L(x) \sqrt{1 + L'(x)^2} dx = \frac{2b}{(a^2 + 1)^{\frac{3}{2}}} \sqrt{a^2 - b^2 + 1}$$

Such that the remaining apple volume is some specific area given by;

$$\begin{aligned} & \int_{-1}^{x_0} \sqrt{1 - x^2} dx + \int_{x_1}^1 \sqrt{1 - x^2} dx + \int_{x_0}^{x_1} (ax + b) dx \\ &= \left(\frac{b}{a^2 + 1} \right) \left(\sqrt{a^2 - b^2 + 1} \right) + \frac{1}{2} \arcsin(x_0) - \frac{1}{2} \arcsin(x_1) + \frac{\pi}{2} \end{aligned}$$

Numerical results indicate that $a = b$ is the minimizer for a value of a which satisfies the volume constraint.

Assuming that each independent integral diverges, we have the following;

$$\lim_{\alpha \rightarrow 1} \frac{\left(\int_0^\infty \alpha^t f(t) dt \right)^2}{\int_0^\infty \alpha^t t f(t) dt} = \lim_{t \rightarrow \infty} f(t)$$

This is related to the idea of Abel regularization of integrals where a similar limit is taken (from below) of an exponential parameter.

If $A(x), B(x)$ are typical Fourier series of the form $\sum_{n=-\infty}^\infty a_n e^{2\pi i n x}, \sum_{n=-\infty}^\infty b_n e^{2\pi i n x}$, then we easily have;

$$\sum_{n=-\infty}^\infty a_n b_n e^{2\pi i n y} = \int_0^1 A(y) B(y - x) dy$$

In much the same way that coefficient-multiplication for Fourier series resembles a Fourier convolution, coefficient-multiplication for a power series resembles a Mellin convolution. (Hadamard)

$$C(z) = \frac{1}{2\pi i} \int_L A(s) B\left(\frac{z}{s}\right) \frac{ds}{s}$$

For a quick application of Abel-Plana, we see a convenient integral form for the following Lambert Series;

$$\sum_{n=1}^\infty \frac{1}{a^n - 1} = \log_a \left(1 + \frac{1}{a - 1} \right) + \frac{1}{2(a - 1)} + \frac{1}{\ln(a)} \int_0^\infty \frac{1}{e^{\frac{2\pi x}{\ln(a)}} - 1} \frac{2a \sin(x)}{a^2 + 1 - 2a \cos(x)} dx$$

With the first two terms being the asymptotic expansion.

We can achieve an interpolation between the determinant ($\alpha = 1$) and the permanent ($\alpha = 0$) of a square matrix by;

$$\sum_{\sigma \in S_n} \cos(\pi k_\sigma \alpha) \prod_i a_{i, \sigma(i)}$$

Where k_σ is the number of transpositions of σ . Further, we can achieve an interpolation between the symmetric polynomials of the eigenvalues as; ($n = 0 \rightarrow \det, n = N - 1 \rightarrow \text{Trace}$)

$$(-1)^n \oint \frac{\det(A_{N \times N} - zI)}{z^{n+1}} dz = \frac{e^{ni\pi}}{2\pi} \int_{-\pi}^\pi \det(A - e^{i\theta} I) e^{-ni\theta} d\theta$$

Zeta double sums;

$$\sum_{m=1}^\infty x^{2m} \sum_{n=1}^\infty (-1)^n (\zeta(2m + 2n) - 1) =$$

$$\frac{x}{2(x^2+1)}\pi \cot(\pi x) + \frac{1}{2}\pi \coth(\pi) \left(\frac{x^2}{x^2+1} \right) - \frac{1}{2} + \frac{1}{2} \frac{x^2}{1-x^2}$$

$$\sum_{m=1}^{\infty} x^{2m} \sum_{l=1}^{\infty} (\zeta(2m+2l) - 1) = \frac{-2}{4(x^2-1)} \left(\pi x \cot(\pi x) + \frac{3}{2}x^2 - 3 - \frac{2}{x^2-1} \right)$$

Immediate consequences of several Mittag Leffler expansions;

$$\sum_{n=1}^{\infty} \frac{1}{n^2+1} \frac{1}{x^2-n^2} = \frac{1}{2x(x^2+1)}\pi \cot(\pi x) + \frac{1}{2}\pi \coth(\pi) \left(\frac{1}{x^2+1} \right) - \frac{1}{2x^2}$$

$N = 85, 86, 87$ are the only integral values for which;

$$\left| \left\{ \zeta \left(\frac{1}{2} + it \right) = 0 ; t < N \right\} \right| = |\{p < N ; \text{primes } p\}| = 23$$

Dodging the 23rd and 24th non-trivial zeros at $\sim 84.7, \sim 87.4$ respectively. Otherwise stated, the n th prime in this region matches closely with the n th zeta-zero;

n	p_n	$\approx \rho_n$
22	79	82.9
23	83	84.7
24	89	87.4
25	97	88.8

Interestingly, the Riemann-Von Mangoldt asymptotic for the number of non-trivial zeros beneath x follows a similar form as Stirling's formula. If $N(t)$ is the typical count of non-trivial zeros with imaginary component less than t , then it is patent;

$$N(2\pi x) - \frac{7}{8} + \ln(\sqrt{2\pi x}) \sim \ln(x!)$$

$$n - \frac{7}{8} + \frac{1}{2} \ln(\rho_n) \sim \ln\left(\frac{\rho_n!}{2\pi}\right)$$

Imagine a large array of N participants in a marble game: in each iteration out of I iterations in total, a pair is randomly selected and the two play an all-or-nothing coin toss with the winner inheriting all the marbles of the opponent. If the crowd is initialized with one marble each, how does the standard deviation of marble ownership vary over time?

If we imagine plotting the variance of the distribution of marbles on a scale from 0 to I/N , where I keeps pace with N at a linear rate, the resulting graph *converges* to the line $y(t) = 2t$. Examining the change in variance between an iteration featuring the pair x_a and x_b in contest:

$$v_{i-1} = \frac{1}{N} \sum_k (x_k - 1)^2 + \frac{1}{N} (x_a - 1)^2 + \frac{1}{N} (x_b - 1)^2$$

$$v_i = \frac{1}{N} \sum_k (x_k - 1)^2 + \frac{1}{N} (0 - 1)^2 + \frac{1}{N} (x_a + x_b - 1)^2$$

$$v_i - v_{i-1} = \frac{2x_a x_b}{N}$$

The probability of a participant remaining at 1 marble (ie not being selected) after a *small* number of k iterations is roughly;

$$P(X_j = 1) \sim 1 - \frac{2k}{N}$$

This it is clear that;

$$\Delta_i v \sim \frac{2i}{N} + O(N^{-2})$$

Hence, after normalizing to units of I/N , we expect $v(k) \sim 2k$.

Take a complex number z in the plane with coordinates specified by a standard Normal distribution. Keeping track of several expectations one has;

$$e = \left(\frac{E\{|z|\}}{E\{\sin(|z|)\}} \right)^2$$

$$\gamma = \ln(2) - 2E\{\ln(z)\}$$

(Assuming $-\pi < \Im \ln(z) < \pi$) More generally, for a particle in the plane driven with components of independent Brownian motion, the expected log of its modulus is;

$$\frac{\ln(2t) - \gamma}{2}$$

$$1 = E\{e^z\}$$

...

Which, from the perspective of numerical simulation, converges quite slowly.

The parameterized integral;

$$\begin{aligned} & \int_0^\infty ((a+x)(b+x))^s dx \\ &= \frac{(ab)^{(s+\frac{1}{2})}}{2} \cdot \int_0^\infty \frac{e^{t(s+1)}}{\sqrt{e^t + \left(\frac{a-b}{2} \cdot \frac{1}{\sqrt{ab}}\right)^2}} dt \\ &= \frac{(ab)^{(s+\frac{1}{2})}}{2} \sum_{n=0}^\infty \binom{-\frac{1}{2}}{n} \left(\frac{a-b}{2} \cdot \frac{1}{\sqrt{ab}}\right)^{2n} \cdot \frac{1}{n - (s + \frac{1}{2})} \end{aligned}$$

A simple bouncing ball released from a height of h_0 which bounces back with a proportion p of its input energy. Will have a settling time of;

$$t_0 + \sqrt{\frac{2h_0}{g} \frac{-2\sqrt{p}}{\sqrt{p}-1}}$$

The n th bounce will occur at;

$$t_0 + \sqrt{2gh_0} \frac{2}{g} \cdot \frac{\sqrt{p}^n - \sqrt{p}}{\sqrt{p}-1}$$

And the maximum height reached as a function of time;

$$h_0 \left(\frac{(t-t_0) \cdot \frac{\sqrt{p}-1}{\sqrt{\frac{2h_0}{g}}} + 2\sqrt{p}}{(1+\sqrt{p})} \right)^2$$

With $t_0 = \sqrt{\frac{2}{g}h_0}$

In umbral notation we may describe the Fokker-Planck equation by;

$$M_{N(\mu,\sigma^2)} \left(-\frac{d}{dx} \right) = e^{\frac{d}{dt}}$$

But what does this mean? Further, the normalization factor for the LIL of $y = \sqrt{-2x \ln(\ln(x))}$ is equivalent to;

$$\exp \left[\frac{1}{2} \sqrt{2\pi x} \cdot N(y \mid \mu = 0, \sigma = \sqrt{x}) \right] = \sqrt{x}$$

Suppose we wish to knot a rectangular strip of paper into a pentagon, of side length 1, following the classic over-and-under knot construction. The minimum dimensions ought to be;

$$\sqrt{\frac{5+\sqrt{5}}{8}} : \frac{1}{4}\sqrt{\frac{5-\sqrt{5}}{2}} + 3 + \sqrt{5}$$

Or the exterior dimensions of alternating 4 trapezoids together made from the ‘bottom half’ of a regular pentagon.

Hilbert’s enumeration of the Rationals as a diagonal path through a grid of each numerator-denominator pair has a generating function;

$$\sum_{n=2}^{\infty} x^{\frac{(n-2)(n-1)}{2}} \sum_{k=1}^{n-1} \frac{n-k}{k} x^k = \sum_{n=1}^{\infty} x^{\frac{(n-1)n}{2}} \left(\int_0^x \frac{t^{(n+1)} - (n+1)t + n}{(t-1)^2} dt \right)$$

Listing out every j th derivative, divided by $j!$, of this function will encompass every rational number eventually.

An equally fun exercise is to formulate the function $R(n)$ which outputs the n th rational number according to the zig-zag pattern.

$$R(x) = \frac{2}{3 + \frac{2x}{M(x)} - M(x)} - 1 ; M(x) = \lfloor \frac{1 + \sqrt{1+8x}}{2} \rfloor + 1$$

Since the number of rationals before the n th is a triangular number, M produces the diagonal on which one increments the numerator and denominator. The procedure of 1) find the pair of triangular numbers j falls between, 2) compute how far along j is from the bottom one, and 3) construct an $\frac{n-k}{k}$ rational is not only used to make the simple formula for R but is also baked into the stated GF .

$$\begin{aligned} P^{\{j\}} &= \sum_{l=0}^j \text{nCr}(j, l) \sum_{n=2}^{\infty} \frac{d^l}{dx^l} \left\{ x^{\frac{(n-2)(n-1)}{2}} \right\} \sum_{k=1}^{n-1} \frac{n-k}{k} \frac{d^{(j-l)}}{dx^{(j-l)}} \{x^k\} \\ P^{\{j\}}(0) &= \sum_{l=0}^j \text{nCr}(j, l) \sum_{n=2}^{\infty} \text{nPr} \left(\frac{(n-2)(n-1)}{2}, l \right) \sum_{k=1}^{n-1} \frac{n-k}{k} \text{nPr}(k, j-l) \{k-j+l=0\} \left\{ \frac{(n-2)(n-1)}{2} - l = 0 \right\} \\ &\sum_{n=0}^{\infty} \text{nPr} \left(j, \frac{(n-2)(n-1)}{2} \right) \sum_{k=1}^{n-1} \frac{n-k}{k} g \left(j, \frac{(n-2)(n-1)}{2} \right) \text{nPr} \left(k, j - \frac{(n-2)(n-1)}{2} \right) \left\{ k - j + \frac{(n-2)(n-1)}{2} = 0 \right\} \\ &\sum_{n=0}^{\infty} \frac{\left(n - j + \frac{(n-2)(n-1)}{2} \right)}{j - \frac{(n-2)(n-1)}{2}} \left\{ j - \frac{(n-2)(n-1)}{2} \geq 1 \right\} \left\{ n-1, j - \frac{(n-2)(n-1)}{2} \right\} \end{aligned}$$

An identical filtration procedure.

The hazards of a blind series regularization;

$$\gamma \sim H_x - \ln(x) \sim \sum_{n=1}^{\infty} \frac{1}{n} \left[e^{-\frac{n}{x}} \cos\left(\frac{n}{x}\right) \right] - \ln(x) \sim -\frac{1}{2} \ln \left(\frac{e^{-\frac{2}{x}} - 2e^{-\frac{1}{x}} \cos\left(\frac{1}{x}\right) + 1}{\frac{1}{x^2}} \right) \sim \ln \left(\frac{1}{\sqrt{2}} \right)$$

A compact limit definitions of the Error Function;

$$2e^{-n} \sum_{k=n+1}^{n+x\sqrt{2n}} \frac{n^k}{k!} \sim_n 2e^{-n} \sum_{k=0}^{n+x\sqrt{2n}} \frac{n^k}{k!} - 1 \sim_n \text{erf}(x)$$

These sorts of (constructed) limits are quite entertaining because of the unexpected use of the Central Limit Theorem.

Take a distribution like Poission in the discrete, or Gamma in the continous, and observe that the sum or the integral is equal to the probability that a sum of N such distributions is less than a helpfully constructed upper bound containing the mean and variance of the distribution in precisely the right

amounts.

Taking $\lim_{n \rightarrow \infty}$ gives rise to the error functions or Normal CDF noted here.

$$\int_0^{N\alpha L + \sqrt{N}\sqrt{\alpha}LQ} \text{Gam}(x, \alpha N, L) dx \sim_N \int_0^{N + \sqrt{N}Q} \frac{(x)^{(N-1)} e^{-x}}{(N-1)!} dx \sim_N \int_{-\infty}^Q \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx$$

Let's examine the Hyperbolic Slot demonstration; an off-center angled rod rotating through a hyperbola-shaped slot. We consider the simplest WLOG case to justify:

1. An initial displacement from the center of rotation by (1,0,0)
2. A tilt of the rod of $\pi/4$ parameterized by the straight line (1,t,t)
3. A rotation about the z-axis defined by $\begin{pmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{pmatrix}$
4. The intersection with the slot/wall occurs on the plane $x = 0$, which provides $\cos(\alpha) - t \sin(\alpha) = 0$
5. Which defines the intersection slot as the curve $z(s) = \cot(s), y(s) = \sin(s) + \cos(s) \cot(s)$, or equivalently $\csc(s)^2 - \cot(s)^2 = 1 \Rightarrow y^2 - z^2 = 1$.
6. A vertical hyperbola; the intersection of a slanted rod with a plane during a circular rotation.

Not mine, but worth restating:

$$\frac{2017}{1475} 2^{\frac{6}{5}} - \pi = 8.8 \times 10^{-12}$$

Also

$$\ln(e^\pi - 20) \frac{90}{103} \sim 1$$

The solution to the Dirichlet problem in the unit disc is given nicely in the form of an integral with the Poisson Kernel;

$$P(\alpha, \beta) = \frac{1 - \beta \cos(\alpha)}{\beta^2 + 1 - 2\beta \cos(\alpha)}$$

The harmonic solution in the rectangle of $\{0 \leq x \leq a; 0 \leq y \leq b\}$, with homogeneous BC on all but the top most edge, where we enforce $g(x)$, can be given by a straightforward Fourier Series.

$$\sum_{n=1}^{\infty} \sin\left(\frac{\pi x}{a} n\right) \frac{\sinh\left(\frac{\pi y}{a} n\right)}{\sinh\left(\frac{\pi b}{a} n\right)} \left(\frac{1}{\frac{a}{2}} \int_0^a \sin\left(\frac{\pi t}{a} n\right) g(t) dt\right)$$

Expanding out the sinh terms gives a suspiciously similar result however;

$$u(x, y) = \frac{1}{a} \int_0^a \sum_{k=0}^{\infty} \left(\hat{P}\left(\frac{\pi x}{a}, \frac{\pi t}{a}, \exp\left(-\frac{\pi b}{a} + \frac{\pi y}{a} - \frac{2\pi b}{a} k\right)\right) - \hat{P}\left(\frac{\pi x}{a}, \frac{\pi t}{a}, \exp\left(-\frac{\pi b}{a} - \frac{\pi y}{a} - \frac{2\pi b}{a} k\right)\right) \right) g(t) dt$$

Where $\hat{P}(x, t, \beta) = P(x - t, \beta) - P(x + t, \beta)$ An infinite series of Poisson Kernels thus establishing a 'Rectangular Kernel'

Perhaps the most beautiful aspect of diagonalization is the generality with which it commutes with functions of the matrix $Eig\{f(A)\} = f(Eig\{A\})$, provided a distinct set of eigenvalues, as will be assumed here.

By Caley-Hamilton, any, potentially infinite, polynomial in A can be reduced to one of degree $n - 1$, such as by polynomial division by the characteristic polynomial of A . Hence, *functional evaluation on matrices is but interpolation of the eigenvalues*.

$$f(A) = \sum_{k=0}^{n-1} A^k c_k; f(\lambda_j) = \sum_{k=0}^{n-1} \lambda_j^k c_k$$

$$f(A) = [I, A, A^2 \dots]^T \text{Vandermonde}\{\lambda_0, \lambda_1, \dots\}^{-1} [f(\lambda_0), f(\lambda_1), \dots]; \lambda_0 \neq \lambda_1 \neq \lambda_2 \dots$$

I find it amusing that under this context, a cyclic permutation matrix, C_n , with an eigenvalue Vandermonde matrix equivalent to the DFT matrix would in fact be computing the FFT of f evaluated at the roots of unity. Of course, we already know that evaluating f on the roots of unity separates components in its Taylor Expansion into residue classes, giving little more than what could be obviously seen;

$$f(C) = \sum_{k=0}^{\infty} C^k c_k = \sum_k^{0 \bmod n} C^k c_k + \sum_k^{1 \bmod n} C^k c_k + \sum_k^{2 \bmod n} C^k c_k \dots = \sum_{m=0}^{N-1} A^m \sum_{k=0}^{\infty} c_{m+kN}$$

One result from the infinitely fascinating Szego Limit Theorems is that, the following limits of determinants of increasingly larger Toeplitz matrices converges to;

$$\lim_{n \rightarrow \infty} \det(T_n)^{1/n} = \exp \left(\frac{1}{2\pi} \int_0^{2\pi} \ln \mathcal{F}_T(t) dt \right)$$

Where $\mathcal{F}_T(t)$ represents a Fourier Series of the matrix coefficients. If we define $\mathcal{A}_A(t)$ to be the average of some infinitely long diagonal trace of a matrix A , then Louisville's Formula is equivalent to;

$$\lim_{n \rightarrow \infty} \det(\Phi_n(t))^{1/n} = \exp \left(\int_{t_0}^t \mathcal{A}_A(t) dt \right)$$

Where $\Phi_n(t)$ is the solution matrix to $\frac{d}{dt} \Phi_n(t) = A_n(t) \Phi_n(t)$. Perhaps there is more to this similarity than a coincidence.

Hercules is studying graph theory before confronting the Hydra and comes across Cayley's formula for the number of labeled trees of n vertices, when he discovers an important connection:

1. The n heads of the Hydra are arranged as the vertices of a tree, with heads connecting between heads but no cross-linking.
2. Each of the heads are distinguishable, and *label-able*, and each time Hercules arrives to the cave, the monster will have transfigured itself into a random labeled tree.
3. One of these heads is the weakest of the beast, and when slain, will immediately kill the Hydra once-and-for-all.
4. So, Hercules takes up the sword and hacks away each of the outermost layer of heads, the vertices furthest from the root and with only one connection.
5. What is the probability that Hercules has slain this weakest and crucial head (over the space of all possible Hydra arrangements)? And, since Greek Gods have an infinite amount of time, what are **Hercules's odds of success as n tends to infinity**?

Hercules stands before an infinitely tall Hydra...

The stacked Cantor Dust is a lovely illustration which, across the x-axis displays a Cantor Set with dimension equal to its vertical y position. The bounding curves give a densely-packed, non-linear appearance that Mandelbrot dubs "theater curtains [that] are topped by the full interval [0,1]". So, what are these bounding curves and what area do they enclose?

By the Fractal Dimension formula, we have that any point along the innermost left bounding curve, formed by removing the 'middle chunk' from the interval follows;

$$\left(x = \frac{1}{r}, y = D = \frac{\ln(2)}{\ln(r)} \right)$$

And we proceed recursively (eliminating the parameter r as we go) by examining only the left hand side of the (symmetric) Cantor Curtains;

$$\begin{aligned} 2^{-\frac{1}{y}} &= x \\ 2^{-\frac{1}{y}} - 4^{-\frac{1}{y}} &= x \\ 4^{-\frac{1}{y}} &= x \end{aligned}$$

$$\int_{\frac{S_\epsilon}{N}}^1 \frac{Nu}{\ln(Nu) - (1 - \epsilon)} \frac{\ln(N)}{N} f'(u) du < [\dots] < \int_{\frac{S_\epsilon}{N}}^1 \frac{Nu}{\ln(Nu) - (1 + \epsilon)} \frac{\ln(N)}{N} f'(u) \frac{1}{N} du$$

And note that the second integral is squeezed between;

$$\int_{\frac{S_\epsilon}{N}}^1 \frac{\ln(N)}{\ln(Nu) + c} u f'(u) du \rightarrow \int_{\sim 0}^1 [\sim 1] u f'(u) du$$

And the first integral vanishes to 0;

$$\left| \int_0^{S_\epsilon} \pi(x) \frac{\ln(N)}{N} f'\left(\frac{x}{N}\right) \frac{1}{N} dx \right| \leq \pi(S_\epsilon) \left| \max f'\left(0 \leq x \leq \frac{S_\epsilon}{N}\right) \right| S_\epsilon \frac{\ln(N)}{N^2} \rightarrow 0$$

Though not practical for a precise asymptotic, it is an entertaining notion to think about selecting a desired function, say $f(x) = x^a$, and computing a prime-Riemann integral which differs by a log factor;

$$\sum_{k \leq N} f\left(\frac{k}{N}\right) \frac{1}{N} \rightarrow_{\text{Related To}} \int_0^x k^a dk = \frac{x^{a+1}}{a+1} \rightarrow_{\text{Prime Log Correction}} \frac{x^{a+1}}{(a+1) \ln(x)} \sim \sum_{p \leq x} p^a$$

But all of these are easily derivable by Abel summation on $\pi(x)$, so the act of defining a ‘prime integral’ is little more than a curiosity.

$$\mathcal{L} \left\{ \frac{1}{\sqrt{\cosh(t)}} \right\} (s) = \sqrt{2} \int_0^{\frac{\pi}{4}} \tan(x)^{(s-\frac{1}{2})} \sec(x) dx = 2^s B \left(3 - 2\sqrt{2}, \frac{s}{2} + \frac{1}{4}, -s + \frac{1}{2} \right)$$

Where $B(x,a,b)$ is the un-normalized incomplete Gamma Function. When $s = 2n + 1 - 1/2, n \in \mathbb{N}$, readily available closed forms exist based on standard results for the middle integral.

$$2^{(2n+1+\frac{1}{2})} B \left(3 - 2\sqrt{2}, n+1, -2n-1 \right) = \sum_{k=0}^n \frac{n \text{Cr}(n,k)}{2k+1} (-1)^{(n-k)} \left(2^{(k+1)} - \sqrt{2} \right)$$

A nice exercise in logs, Sterling’s formula, and the Hyperfactorial is;

$$\prod_{n=2}^{\infty} \frac{n!}{\sqrt{2\pi n} \left(\frac{n}{e}\right)^n} \left(1 - \frac{1}{n}\right)^{\frac{1}{12}} = (2\pi)^{\frac{3}{4}} \frac{e^{-\frac{11}{12}}}{A^2}$$

Deduced from nothing more than logs and limits $\lim_{N \rightarrow \infty}$;

$$\sum_{n=2}^N \sum_{k=2}^n \ln(k) - \ln(2\pi) \frac{(N-1)}{2} - \frac{1}{2} \ln(N!) - \sum_{n=2}^N n \ln(n) + \frac{N(N+1)}{2} - 1$$

$$\left(N + \frac{1}{2}\right) \ln \left(\sqrt{2\pi N} \left(\frac{N}{e}\right)^N \left(1 + \frac{1}{12N}\right) \right) - \ln(2\pi) \frac{(N-1)}{2} - 2 \ln \left(AN^{\frac{(6N^2+6N+1)}{12}} e^{-\frac{N^2}{4}} \right) + \frac{N(N+1)}{2} - 1$$

And hence the appearance of A , the Glaisher–Kinkelin constant.

$$\frac{1}{|P^+|} \int_{P^+} f(x) df_c = \frac{|P_0|}{2} \frac{1}{1-x}$$

The averaged integral over the space of all possible polynomial functions with positive coefficients diverges up to the size of the space of one of the coefficients, times a known function.

If we try to reason with the idea of integrating over all possible functions, the restriction to polynomials makes the ‘measurement’ quite easy; take the limit as the degree, k , and coefficient upper bound, N , go to infinity, dividing out by the $k+1$ coefficients we choose, each covering a linear space of N .

$$\lim_{k \rightarrow \infty} \frac{1}{N^{k+1}} \left(\int_0^N \right)^{k+1} \left(\sum_{n=0}^k c_n x^n \right) dc_0 dc_1 \dots dc_{k+1}$$

$$\lim_{k \rightarrow \infty} \sum_{n=0}^k x^n \frac{1}{N^{k+1}} \left(\frac{N^2}{2} - \frac{0^2}{2} \right) \left(\int_0^N \right)^k dc_0 dc_1 \dots dc_{n-1} dc_{n+1} \dots dc_{k+1}$$

$$\lim_{N \rightarrow \infty} \frac{N}{2} \sum_{n=0}^{\infty} x^n \text{ " = " } \frac{|P_0|}{2} \frac{1}{1-x}$$

Similarly, any class of functions which permits such a parametrization can be ‘integrated’ over its set of parameters. Suppose we take S to be the class of sinusoids, of any phase and frequency and compute;

$$\frac{1}{|S|} \int_S f^2(x) df_c = \frac{1}{N} \int_0^N \sin(wX)^2 dw = \frac{1}{\pi} \frac{1}{N} \int_0^N \int_0^\pi \sin(wX - a)^2 dw da = \frac{1}{2}$$

The generating function for the power sums of natural numbers up to m is given by;

$$\sum_{n=0}^{\infty} x^n \left(\sum_{k=0}^m k^n \right) = \int_0^1 \frac{(t^{-x})^{(m+1)} - 1}{t^{-x} - 1} dt$$

That this integral expression is even *differentiable* at $x = 0$ is not so easily seen. To explore the division in integral behavior as x approaches 0 from above and below we write in the following fashion;

$$\int_0^{\infty} (\operatorname{sgn}(y-1) \operatorname{sgn}(x) + 1) \frac{y^{(m+1)} - 1}{y - 1} \left(\frac{\operatorname{sgn}(x)}{2x} \right) y^{(-\frac{1}{x}-1)} dy$$

Which contains the nugget of the harmonic numbers $\int_0^1 \frac{y^{(n+1)} - 1}{n-1} dy$.

A most delightful result concerning alternating sums of the Trigamma function;

$$\sum_{k=0}^{2^{(N-2)}-1} (-1)^k \left(\psi_1 \left(\frac{4k+1}{2^N} \right) - \psi_1 \left(\frac{4k+3}{2^N} \right) \right) = \pi^2 \cdot \sqrt{2}^{4N-7}$$

When cooking up such a formula of rational arguments to a polygamma function we must of course use the corresponding Gauss Theorem. Fortunately we have;

$$\sum_{k=0}^{2^{(N-p)}-1} (-1)^k \left(\sin \left(2\pi m \frac{4k+1}{2^N} \right) - \sin \left(2\pi m \frac{4k+3}{2^N} \right) \right)$$

And so the non-elementary component of the sum is eliminated! Leaving behind;

$$\frac{\pi^2}{2} \sum_{k=0}^{2^{(N-2)}-1} (-1)^k \left(\csc^2 \left(\pi \frac{4k+1}{2^N} \right) - \csc^2 \left(\pi \frac{4k+3}{2^N} \right) \right) = \frac{\pi^2}{2} 2^{\frac{(4N-5)}{2}}$$

Giving the above expression.